



PROJECT REPORT

Interactive Dashboard Design for Formula 1 2025 Season Analysis

Course: Data Visualization – IT4023E

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Abstract

This report presents the design and implementation of an interactive Power BI dashboard for analysing the 2025 Formula 1 World Championship. The project integrates multiple related datasets, including drivers, constructors, race calendar, qualifying results, race results, and fan voting, into a coherent analytical model. The dashboard is organised into Overview, Teams, Track, and Drivers pages, allowing users to explore season-level outcomes, team performance, circuit characteristics, and individual driver consistency.

The report focuses not only on dashboard functionality but also on the application of data visualization principles covered in the course. Visual encodings are selected according to data types and analytical tasks, with emphasis on position, length, colour usage, proportional ink, interaction, and storytelling. Through coordinated slicers, KPI cards, bar charts, line charts, scatter plots, tables, and supporting visual elements, the dashboard transforms fragmented racing data into a clear and interpretable analytical tool.

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Introduction

1.1 Background and Motivation

Formula 1 (F1) is widely recognized as one of the most technologically advanced and data-intensive sports. During a season, large volumes of data are generated from race weekends, including drivers, constructors, circuits, qualifying sessions, and race results. While these data contain rich information about performance and competitive dynamics, they are typically stored in separate tables and published as raw statistics, which are difficult to interpret without appropriate visual support.

Modern sports analytics increasingly relies on data visualization to turn complex datasets into meaningful and actionable insights. Interactive dashboards allow users to quickly identify trends, compare entities, and explore performance patterns from multiple perspectives. In this context, a well-designed visualization system for F1 can support different audiences such as fans, students, analysts, and researchers in understanding the 2025 championship more effectively.

1.2 Problem Definition

Although F1 data are publicly available from various sources, they are fragmented across multiple datasets and usually presented as static tables or simple charts. Such representations make it hard to answer higher-level questions, for example: how consistently a driver performs across tracks, how teammates contribute to team points, or how circuit characteristics relate to race outcomes. Furthermore, many existing visualizations are not explicitly designed with data visualization principles in mind, which may lead to misleading encodings, ineffective color choices, or unnecessary visual clutter.

This project addresses the need for an interactive, principle-driven dashboard that consolidates F1 2025 season data into a coherent analytical environment. The focus is not only on building a functional Power BI dashboard, but also on systematically applying the visualization concepts covered in the course (data types, visual channels, pre-attentive processing, proportional ink, interaction, and storytelling). The written report will explicitly link design decisions in the dashboard to these theoretical principles.

1.3 Project Goals and Scope

The overall goal of the project is to design and implement an interactive dashboard for analysing the 2025 Formula 1 World Championship using Power BI. The dashboard aims to transform raw racing data into clear, interpretable visualizations that support both exploratory analysis

and explanatory communication for non–expert users.

Specifically, the project pursues the following objectives:

- Collect and integrate multiple F1 datasets for the 2025 season, including drivers, constructors, race calendar, qualifying results, race results, and fan voting.
- Clean and preprocess the data to ensure consistency of identifiers (drivers, teams, circuits) and to support reliable joins between tables.
- Design a relational data model in Power BI that connects entities such as drivers, teams, tracks, and race events.
- Build an interactive dashboard with multiple pages that provide high–level overviews as well as detailed views for teams, drivers, and circuits.
- Apply core data visualization principles from the course (e.g., expressiveness and effectiveness, graphical perception, color usage, proportional ink, interaction and storytelling) and document how they are used in each dashboard page.

Dataset Overview

2.1 Data Sources and Structure

The dataset used in this project focuses on the 2025 Formula 1 World Championship. It was compiled from publicly available F1 resources and is organised into several related tables: *Calendar*, *Drivers*, *Teams*, *QualifyingResults*, *RaceResults*, and *DriverOfTheDayVotes*. Together, these tables describe race events, circuits, drivers, constructors, qualifying performance, race outcomes, and fan engagement.

The data are stored in a relational structure, where common keys such as driver identifiers, team names, circuit codes, and race rounds are used to establish relationships across tables. This structure supports the construction of a unified analytical model in Power BI and enables cross-filtering between different dimensions of the championship.

2.2 Main Data Categories and Types

For the purpose of visualization design, the attributes in the dataset can be grouped into four main categories:

- **Driver information** (from *Drivers*): nominal attributes such as driver name, team, nationality, and place of birth; ordinal or count attributes such as number of championship titles and podiums; quantitative measures such as total career points and date of birth (used to derive age).
- **Constructor information** (from *Teams*): nominal attributes such as team name, chassis, and power unit supplier; quantitative attributes such as historical team points, number of wins, and podiums.
- **Race performance** (from *QualifyingResults* and *RaceResults*): ordinal attributes such as grid position and finishing position; quantitative attributes such as lap times, race time, points scored, and fastest lap; categorical status variables (e.g., finished, DNF).
- **Circuit information** (from *Calendar*): nominal attributes such as Grand Prix name, country, and city; quantitative attributes such as circuit length, race distance, number of laps, number of turns, and number of DRS zones; quantitative lap record times.

Explicitly distinguishing nominal, ordinal, and quantitative variables is important because it determines which visual encodings (position, length, color hue, color value, shape, etc.) are appropriate for each attribute. This classification will be referenced in later chapters when discussing the choice of chart types and visual channels for the dashboard.

2.3 Overview of Data Preparation

Before building the dashboard, several preprocessing steps were performed. Minor inconsistencies in column names, text formatting, and categorical values were corrected so that drivers, teams, and circuits match across tables. Data types were converted appropriately in Power Query (e.g., numeric, date, categorical), and a small number of calculated measures were created to support metrics such as cumulative points and driver–team comparisons.

A simple star–like data model was then constructed in Power BI, linking fact tables (qualifying and race results) with dimension tables (drivers, teams, calendar). This model provides the foundation for the interactive visualizations described in the subsequent chapters.

All cleaning and transformation steps were implemented in Power Query, and season-level measures such as cumulative points, podium rates, and grid-finish gain/loss were defined using DAX. These modelling choices enabled the dashboard to remain responsive while supporting interactive filtering across pages.

Dashboard Design and Visual Analysis

3.1 Overview Page

Overview Page Description. The Overview page (Figure 3.1) serves as the main entry point of the F1 dashboard, offering a macro-level summary of the season before users proceed to more specialised pages such as Teams, Drivers, and Track. The layout is structured to first present key season outcomes, including champions and headline KPIs, followed by rankings for drivers and teams. It then transitions into relationship analysis and concludes with distributions of key performance metrics.

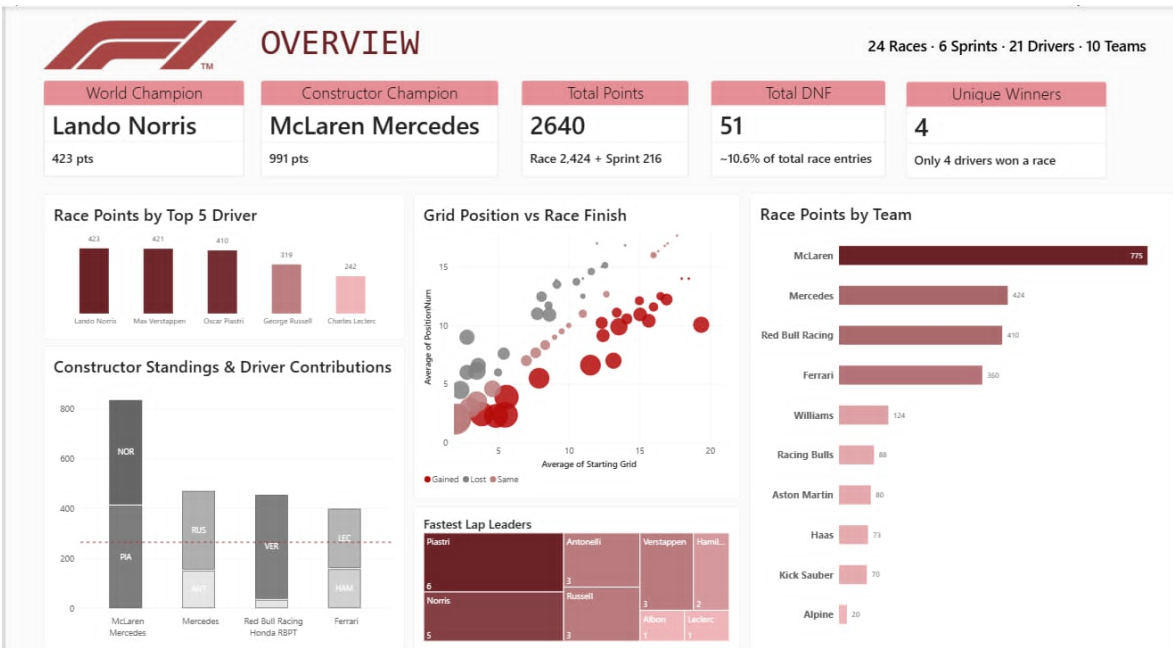


Figure 3.1. Overview Page – Season-level summary of drivers and constructors.

The page is designed to summarise the winners of both the drivers’ and constructors’ championships along with their respective point totals. It also provides high-level indicators such as total points scored across the season, the number of DNFs, and the count of unique race winners. In addition, it illustrates how points are distributed among top drivers and teams, examines how effectively drivers convert starting grid positions into race finishes, and highlights the distribution and concentration of fastest lap performances.

Interactivity is enabled through a Driver slicer and a Team slicer. The Driver slicer filters driver-related visuals, such as the Grid vs Finish scatter plot and Fastest Lap Leaders treemap,

or highlights selected drivers. The Team slicer filters team-related visuals, including Race Points by Team and Constructor Standings, and restricts other visuals, such as scatter plots and treemaps, to drivers belonging to the selected teams.

Visual 1: World Champion Card. This visual is implemented as a KPI card designed to present a season outcome summary through filter-aware aggregation.

The card displays the name of the world champion driver alongside their total championship points. As a KPI card, it does not utilise axes. The design employs a neutral background with slightly emphasised text, where the champion's name is shown in a larger font and the points value in a smaller supporting format. No legend is required. The visual responds to global season filters, such as subsets of rounds, but remains unaffected by the Driver slicer to preserve the global definition of the champion. The main measures used are `Champion Name`, representing the driver with the highest total points, and `Champion Points`, defined as the maximum `Total Points` across all drivers.

The visual answers the question of who the drivers' world champion is and how many points they scored.

A KPI card provides the most direct and concise way to present a single outcome without unnecessary visual elements. More complex representations such as ranking charts are intentionally deferred to subsequent visuals, allowing this card to communicate the key result at a glance.

Visual 2: Constructor Champion Card. This visual also adopts a KPI card format to summarise the constructors' championship outcome.

The card presents the name of the constructor champion along with total constructor points. Similar to the previous KPI, no axes are used. The design may incorporate subtle team colour accents or logos to enhance brand recognition, while maintaining consistency in typography hierarchy with the first visual. No legend is included. The card responds to global season filters but does not change when a single team is selected via the Team slicer, ensuring the championship result remains global. The measures used are `Constructor Champion Name` and `Constructor Champion Points`.

It identifies which team won the constructors' championship and the number of points they achieved.

Since the constructors' champion represents a single key outcome, a KPI card ensures clarity and prominence. Maintaining visual consistency with the drivers' champion card allows users to interpret both results quickly and cohesively.

Visual 3: Total Points Card (Race + Sprint). This visual is implemented as a KPI card to display a season-level aggregation of points.

The card shows the total number of points distributed throughout the season, combining both race and sprint points. As with other KPI cards, no axes or legends are required, and the design remains neutral without additional colour encoding. When filters such as Team or Driver slicers are applied, the value dynamically recalculates to reflect the selected subset. The primary measure is `Total Points = SUM(Race Points + Sprint Points)`.

It communicates the total number of points distributed across the season and, when filtered, the contribution of specific subsets such as individual teams.

A single aggregated value is sufficient to convey the scale of the points system, making the KPI card an appropriate and efficient choice, while more detailed breakdowns are handled in other visuals.

Visual 4: Total DNFs Card. This KPI card is used as a reliability indicator, summarising DNF statistics.

The card displays the total number of DNFs across all race entries and may optionally include a DNF rate calculated as $\text{DNF Rate} = \text{Total DNFs} / \text{Total Starts}$. No axes or legends are required. A subtle warning colour may be applied to reflect reliability concerns. The visual responds to both Team and Driver slicers, allowing users to view DNFs for specific entities. The measures used are Total DNFs and, if included, DNF Rate.

It addresses the overall level of retirements in the season, expressed both as an absolute count and optionally as a percentage.

Reliability metrics are effectively summarised through a single value, making the KPI card an appropriate choice, while more detailed patterns are reserved for other dedicated visuals.

Visual 5: Unique Winners Card. This KPI card functions as a diversity indicator within the season.

The visual displays the number of distinct drivers who have achieved at least one race win. No axes or legends are required, and the design remains neutral. Depending on the measure definition, applying the Team slicer may limit the count to winners within a specific team. The main measure is $\text{Unique Winners} = \text{DISTINCTCOUNT}(\text{Drivers with at least one win})$.

It evaluates whether race wins are concentrated among a small number of drivers or distributed across a broader group.

A single numeric value sufficiently conveys this insight, allowing the KPI card to present the information clearly while detailed driver-level breakdowns are shown elsewhere.

The main KPI band is shown in Figure 3.2.

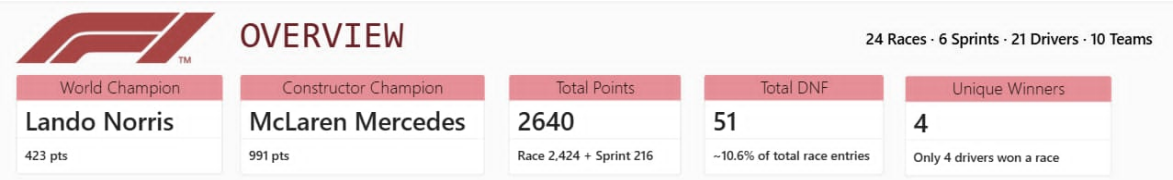


Figure 3.2. Overview Page KPI cards showing the season champion, constructor champion, total points, DNFs, and unique winners.

Visual 6: Race Points by Top 5 Drivers. This visual uses a bar chart to perform ranking and comparative analysis among drivers.

Drivers are plotted along the X-axis, while the Y-axis represents Total Race Points. Colour encoding distinguishes teams using a qualitative palette, and all bars share equal width.

A legend identifies team colours. The visual interacts with both Driver and Team slicers, allowing filtering or highlighting of specific drivers or teams. The main measure is `Total Race Points (Driver)`.

It identifies the top five drivers by race points and illustrates the gaps between them.

Bar charts are well suited for ranking tasks, enabling clear comparison of magnitudes and straightforward interpretation of differences, unlike alternatives such as pie charts.

The resulting ranking chart is shown in Figure 3.3.

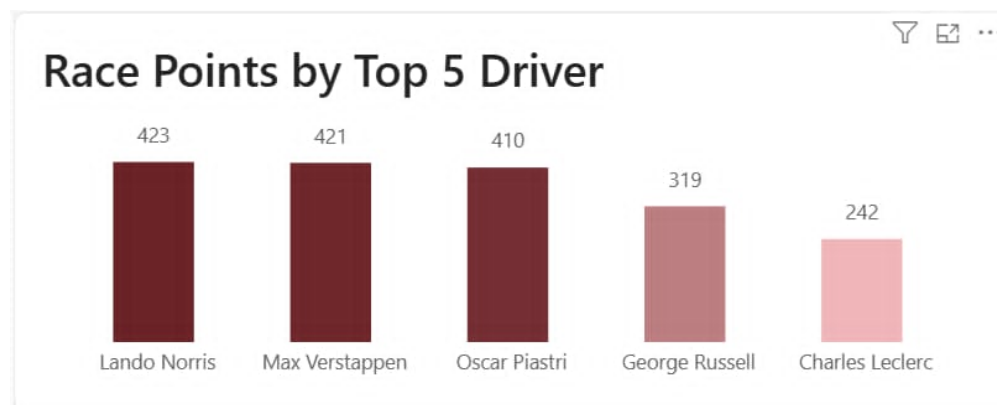


Figure 3.3. Top five drivers ranked by race points on the Overview page.

Visual 7: Grid Position vs Race Finish (Average per Driver). This visual is implemented as a scatter plot to analyse relationships between variables.

The X-axis represents `Average Starting Grid`, while the Y-axis shows `Average Finish Position`. Colour encodes the `GridResult` category (Gained, Lost, Same), and marker size reflects `Total Points`. A legend explains the categories, and tooltips provide additional driver and team details. The visual responds to Team slicers by filtering drivers and allows highlighting via the Driver slicer. Measures include `Average Starting Grid`, `Average Finish Position`, `GridResult Category`, and `Total Points`.

It examines how effectively drivers convert starting positions into finishing positions and identifies those who consistently gain or lose places.

Scatter plots are appropriate for analysing relationships between two quantitative variables, while colour and size add contextual layers without requiring additional visuals.

This relationship view is illustrated in Figure 3.4.

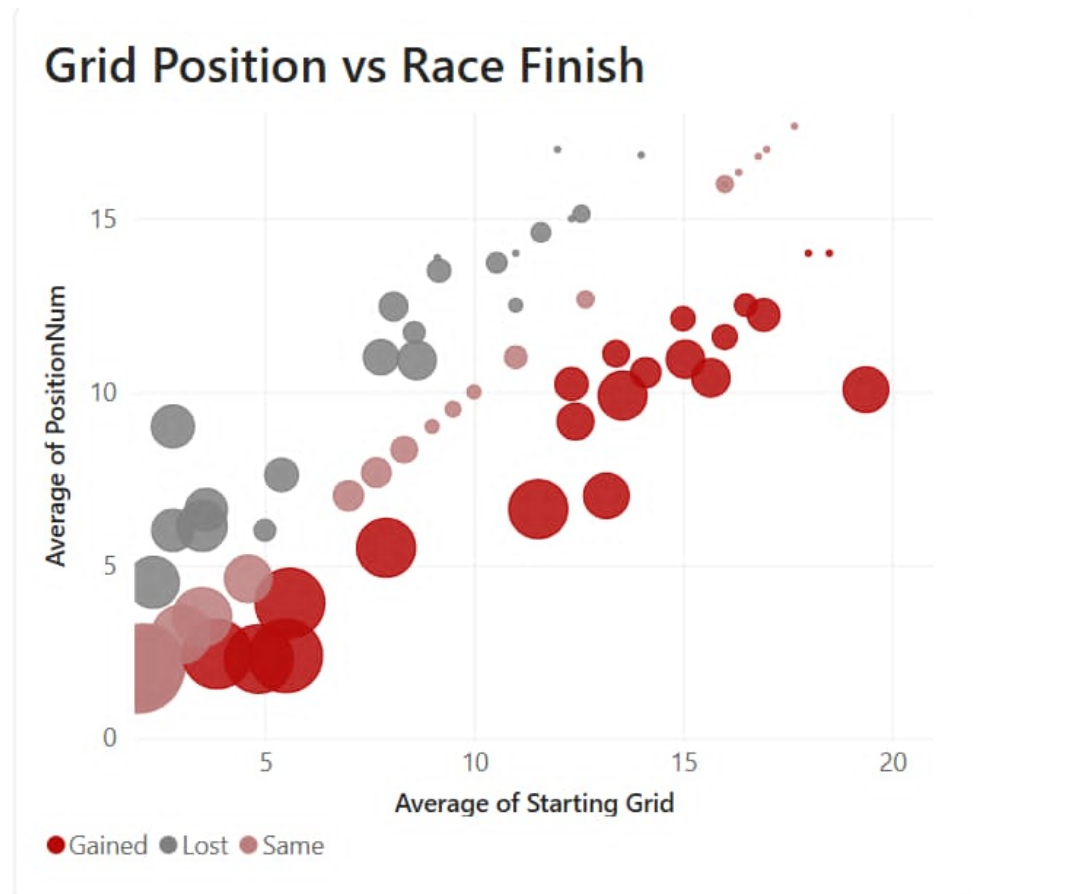


Figure 3.4. Overview scatter plot comparing average starting grid position and average race finish.

Visual 8: Race Points by Team. A horizontal bar chart is used to compare and rank teams.

The X-axis represents Total Race Points (Team), and the Y-axis lists teams. Colours correspond to team identities and remain consistent across the dashboard. Bar length conveys magnitude, and a legend may be omitted if labels are sufficient. The Team slicer enables filtering or highlighting. The primary measure is Total Race Points (Team).

It shows which team scored the most race points and how points are distributed among teams.

Horizontal bars improve readability for longer team names while maintaining clear comparison and ranking.

The team ranking is shown in Figure 3.5.

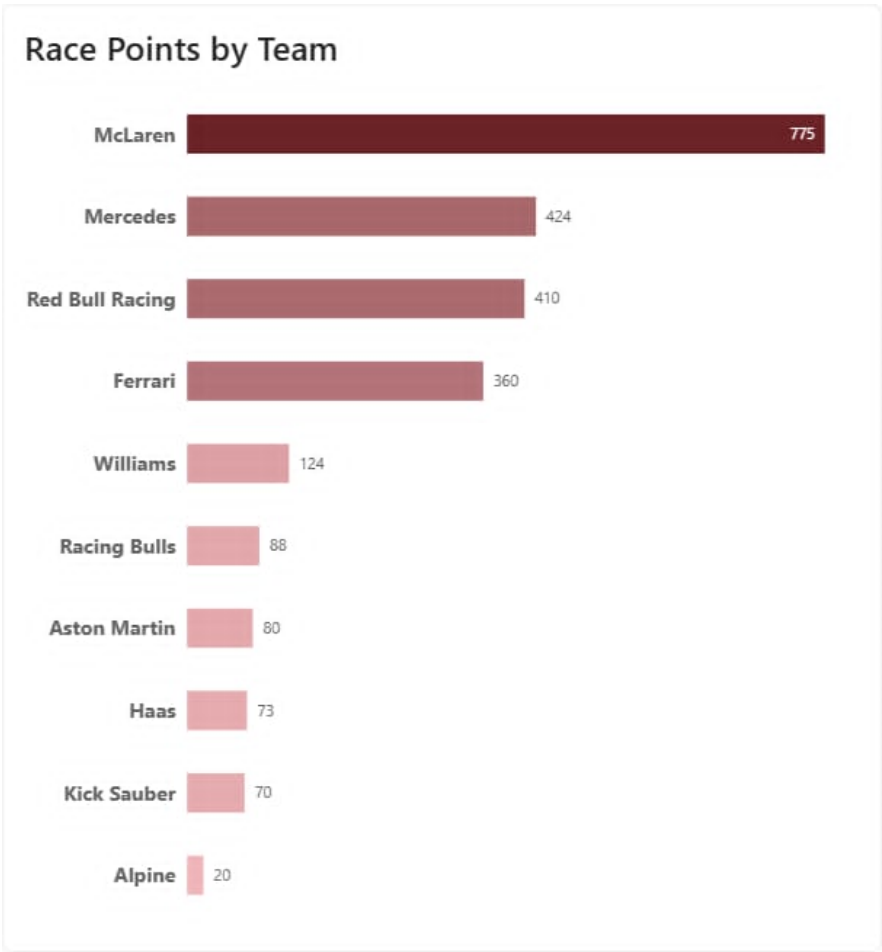


Figure 3.5. Team race points displayed as a horizontal ranking chart.

Visual 9: Constructor Standings and Driver Contributions. This visual is a stacked column chart with an optional reference line, combining ranking and part-to-whole analysis. Teams are displayed along the X-axis, with Total Constructor Points on the Y-axis. Each column is segmented by driver contributions, indicated through colour encoding. A reference line may be added as a benchmark. A legend identifies drivers. When a single team is selected, the visual effectively becomes a breakdown of that team’s points by driver. Measures include Total Constructor Points (Team) and Driver Points Contribution.

It explains both total constructor points per team and the contribution of individual drivers. Stacked columns allow simultaneous interpretation of total performance and internal distribution, eliminating the need for separate visuals.

The constructor contribution view is shown in Figure 3.6.

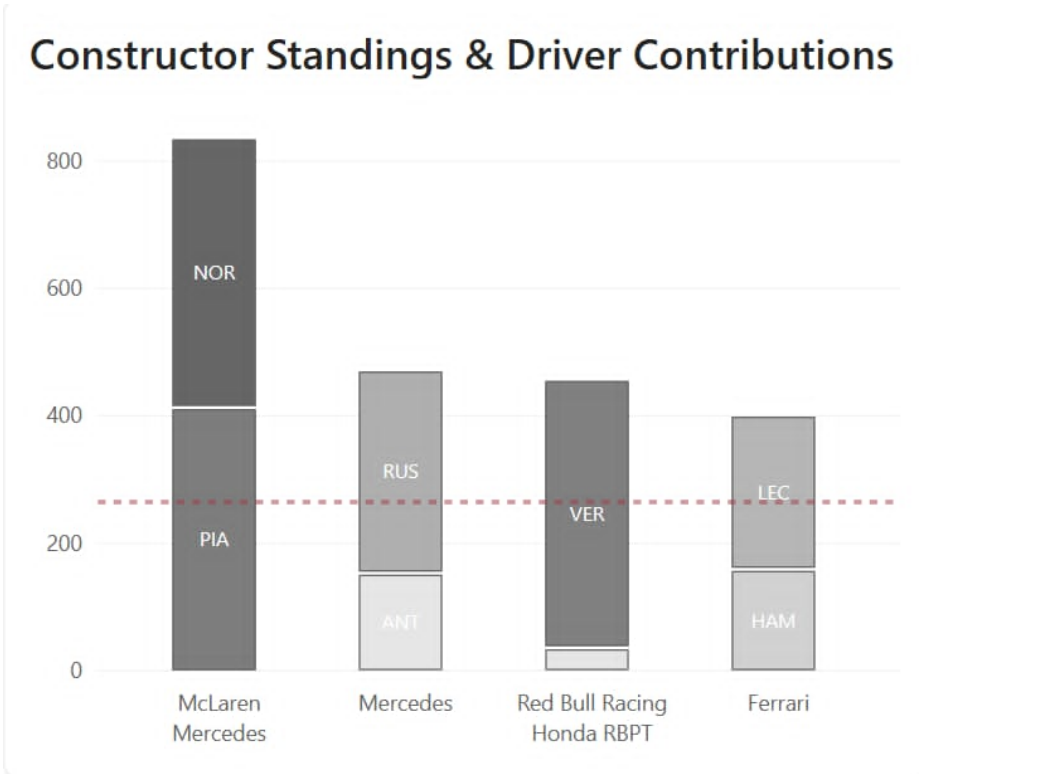


Figure 3.6. Constructor standings with stacked driver contributions and a benchmark reference line.

Visual 10: Fastest Lap Leaders (Treemap). A treemap is used to analyse distribution and dominance patterns.

Drivers form the categories, while tile area represents `Fastest Laps`. Colour follows a single-hue sequential scale, with darker shades indicating higher values. Labels are typically displayed directly on tiles, supported by tooltips for exact values. The Team slicer filters the included drivers. The measure used is `Fastest Laps (Driver)`.

It identifies drivers leading in fastest laps and evaluates whether these performances are concentrated or widely distributed.

Treemaps efficiently display many categories within a compact space and make dominance patterns immediately visible, while detailed rankings can be obtained from other chart types.

The treemap used for fastest lap leaders is shown in Figure 3.7.

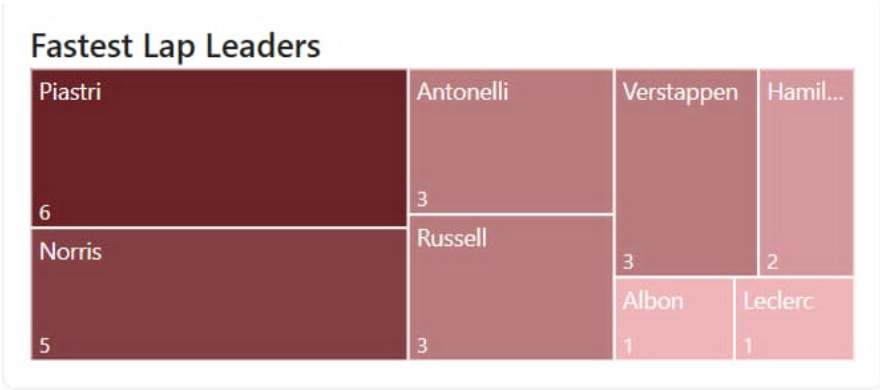


Figure 3.7. Fastest Lap Leaders treemap showing the distribution of fastest laps by driver.

3.2 Teams Page

Teams Page Description. The Teams page (Figure 3.8) is designed as a constructor-centric view of the season. It concentrates on average team performance, the frequency of podium finishes, the contribution of individual drivers across tracks, and the relationship between current performance and historical success. The Team slicer functions as the primary interaction mechanism and transforms the page into a coordinated set of views for the selected constructor.

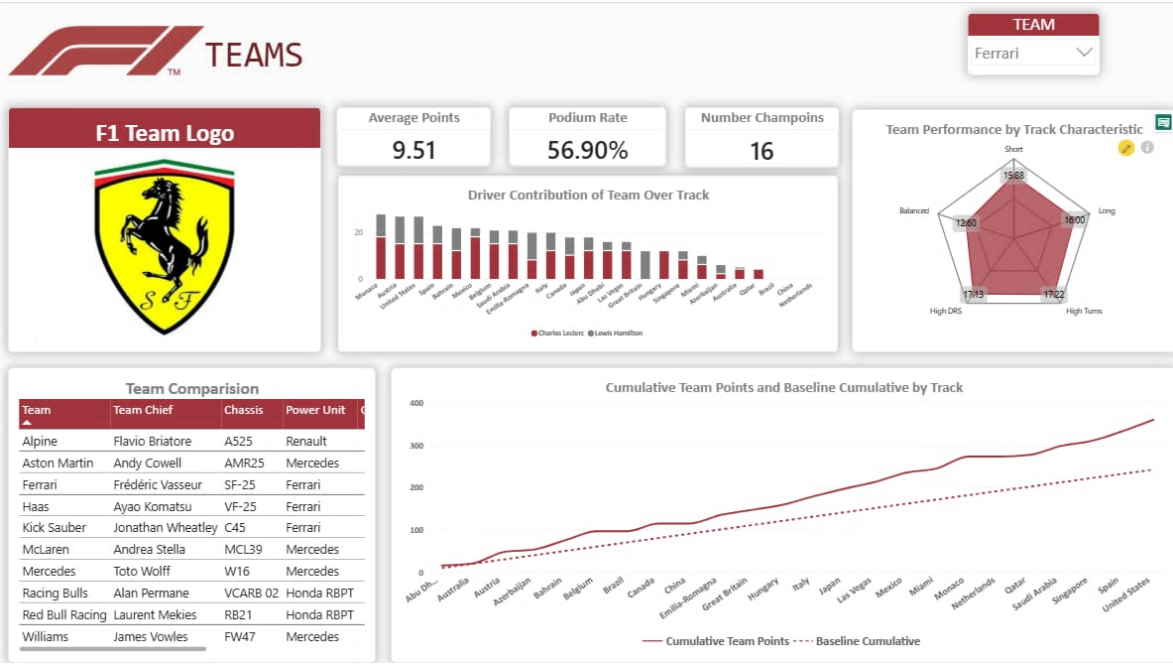


Figure 3.8. Teams Page – Constructor-level performance and driver contributions.

The page is intended to provide a high-level profile for each team by presenting its logo, average points, podium rate, and number of titles. It also shows how team performance varies across different track characteristics and examines how individual drivers contribute to the team’s results at each track. In addition, technical and historical information, including chassis, power unit, team chief, and total races, is presented alongside a comparison between cumulative team points and a baseline expectation over the season.

Interactivity on this page is centred on the Team slicer, which is configured as a single-select control and updates all visuals displayed on the page. Where applied, a Position filter further restricts charts to selected finishing ranges, such as podium finishes or top-10 results.

Visual 1: F1 Team Logo. This visual is implemented as an image visual and is used to display team identity and provide contextual reference.

The content consists of the official logo of the selected team, such as Ferrari, Red Bull Racing, or Mercedes. As an image visual, it does not require X-axis or Y-axis fields. Its colour and size are determined entirely by the logo itself, and no data is encoded visually through these properties. A legend is unnecessary. The Team slicer controls the displayed logo by updating the selected team, and the main field used is `Teams[Team Logo URL]` or an equivalent image path.

The visual identifies which team is currently being analysed and how that team is visually represented within the dashboard.

A logo image is the most intuitive method for identifying a team and linking the dashboard to official F1 branding. It also establishes immediate visual context for the quantitative visuals that follow.

Visual 2: Average Points per Race (KPI). This visual is presented as a KPI card and functions as an average performance indicator.

The card displays `Avg Points` per race for the selected team. Because it is a KPI card, axes are not applicable. The design uses a neutral or slightly emphasised colour treatment without encoding additional variables, and no legend is required. The Team slicer determines which constructor's average is shown, while the Position filter, if applied, can restrict the points included before averaging. The main measure is `Avg Points = AVERAGE(RaceResults[Points])` for the selected team.

This KPI addresses the question of how many points, on average, the selected team scores per race.

A single KPI value is sufficient to communicate team efficiency per race, and a full chart is not necessary for this purpose. The card also serves as a quick benchmark when users compare teams through the Team slicer.

Visual 3: Podium Rate (KPI). This visual uses a KPI card to show a frequency or ratio indicator.

The card displays the percentage of races in which the team achieved at least one podium finish, defined as positions 1 to 3. No axes are used, and the colour treatment remains neutral or lightly highlighted to reflect success. A legend is not needed. The Team slicer determines which team's podium rate is calculated. The main measures are `Podium Finishes = COUNTROWS(FILTER(RaceResults, Position <= 3))`, `Total Races`, and `Podium Rate = DIVIDE(Podium Finishes, Total Races)`.

The KPI shows how frequently the selected team reaches the podium relative to the total number of races.

Podium frequency is most naturally interpreted as a single percentage value, and the card format keeps the result clear and compact. This also supports quick comparison across teams

when the Team slicer is changed.

Visual 4: Number of Constructors' Championships (KPI). This visual is a KPI card used to provide historical context.

The card presents the number of constructors' titles won historically by the selected team. Since it is a KPI card, no axes are involved. The colour styling remains neutral, although it may be formatted to emphasise team legacy, and no legend is needed. The Team slicer determines which constructor's historical title count is shown, while race-level filters do not affect the result because it is treated as a historical constant. The main measure is `Number Championships = SUM(Teams[World Championships])`.

It answers the question of how successful the selected team has been historically in terms of constructors' championships.

Because historical titles are represented by a single scalar value for each team, a simple card is sufficient and avoids unnecessary visual clutter. It also allows current performance to be interpreted alongside historical legacy without increasing complexity.

Visual 5: Team Performance by Track Characteristic (Radar Chart). This visual is implemented as a radar chart for multi-dimensional comparison across track types.

The axes of the radar chart represent track characteristics, namely Short, Long, High Speed, High Turns, and Mixed. The radial value corresponds to `Avg Points` for each characteristic. Colour is used to distinguish teams if more than one is compared, or to represent the selected team alone when the Team slicer is applied in single-select mode. Distance from the centre encodes magnitude, so higher average points appear farther from the centre. A legend is included when multiple teams are displayed. The Team slicer typically focuses the radar chart on the chosen team. The main measure is `Avg Points by Characteristic = AVERAGE(RaceResults[Points])` grouped by track characteristic.

The chart shows on which types of tracks, including short, long, high-speed, high-turns, and mixed circuits, the selected team performs strongly or weakly.

A radar chart offers a compact way to present performance across multiple dimensions and makes the team's pattern of strengths and weaknesses visible at a glance. For a small and fixed set of characteristics, this format summarises the pattern effectively, while more detailed values remain accessible through tooltips or supporting visuals.

The track-characteristic profile is shown in Figure 3.9.

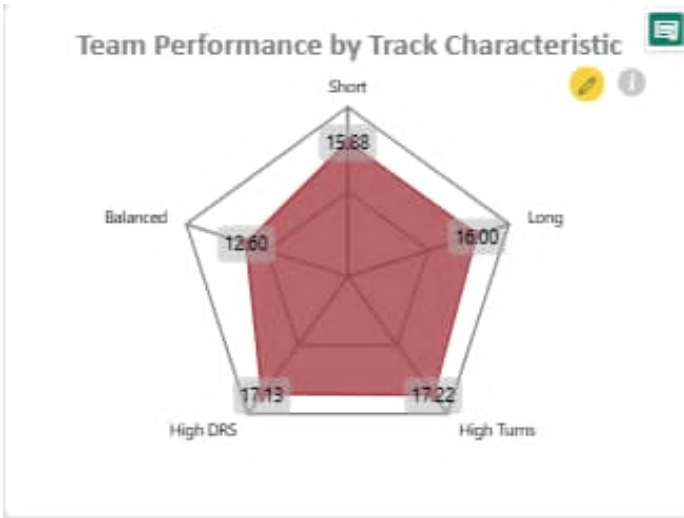


Figure 3.9. Radar chart summarising team performance across track characteristics.

Visual 6: Driver Contribution of Team over Track (Clustered Column Chart). This visual is a clustered column chart used for comparing drivers within a team and analysing contributions by track.

The X-axis represents *Track*, while the Y-axis shows *Points* for the selected team at each track. Colour distinguishes drivers, with each driver assigned a separate colour. Size is not encoded beyond the height of each column, and a legend maps colours to drivers. The Team slicer restricts the visual to a single constructor, and the Position filter may be used to limit the results to certain finish ranges, such as podium positions. The main measure is $\text{Driver Contribution} = \text{SUM}(\text{RaceResults}[\text{Points}])$ grouped by *Track* + *Driver* for the selected team.

The chart identifies which driver contributed more points to the team at each track and which tracks generated the highest total points.

Clustered columns support direct comparison between teammates at the same track because the bars are placed on a common scale side by side. A stacked chart would make direct driver-to-driver comparison less clear, whereas clustering preserves it effectively.

The team identity, KPI, and driver contribution area is shown in Figure 3.10.



Figure 3.10. Teams Page identity, KPI cards, and driver contribution chart for the selected constructor.

Visual 7: Team Comparison Table. This visual is presented as a table and is used for detailed attribute display involving both text and numeric reference information.

The table includes the columns Team, Team Chief, Chassis, Power Unit, Grand Prix Entered, Highest Grid Position, and Team Points. Because of the tabular format, X-axis and Y-axis fields are not applicable. Colour usage is minimal and generally limited to headers or emphasis, and no legend is required. The Team slicer can be used to highlight or filter to the selected team's row. The main measures and fields are `Grand Prix Entered = DISTINCTCOUNT(Calendar[Race])`, `Highest Grid Position = MIN(RaceResults[Grid])`, and `Team Points = SUM(RaceResults[Points])`.

The table provides the key technical and organisational attributes of each team together with its overall points record.

A table is the most appropriate format for presenting a mixture of textual and numerical information with exact values. It complements the charts by allowing precise lookup of details such as chassis, power unit, and other team attributes.

Visual 8: Cumulative Team Points vs Baseline (Line Chart). This visual is a two-line chart used for time-series analysis and comparison against a baseline.

The X-axis represents Track or Round in race order, while the Y-axis shows both cumulative team points for the selected constructor and the baseline cumulative points. Colour distinguishes the actual cumulative series from the baseline series, and size is not used. A legend identifies “Cumulative Team Points” and “Baseline Cumulative.” The Team slicer determines which team's cumulative performance is shown, while a Round filter, if available, can restrict the visible segment of the season. The main measures are `Team Points = SUM(RaceResults[Points])`, `Cumulative Team Points = CALCULATE([Team Points], FILTER(ALL(Calendar), Calendar[Round] <= MAX(Calendar[Round])))` and `Baseline Cumulative = Target Avg Points * Current Round`.

The chart addresses whether the team is accumulating points faster or slower than the expected baseline over the course of the season, and in which stages performance accelerates or slows.

Line charts are well suited to showing changes over race rounds, and the use of two lines allows divergence from the baseline to be identified easily. This makes periods of stronger performance, where the actual line rises above the baseline, and weaker performance, where it falls below, immediately visible.

The supporting table and cumulative comparison are shown in Figure 3.11.

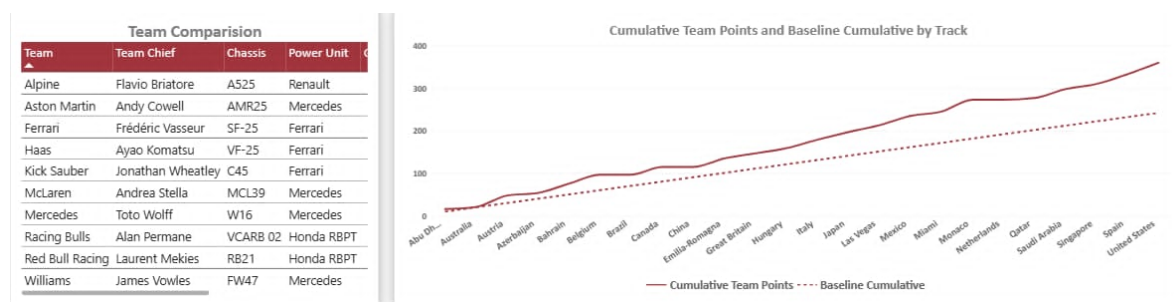


Figure 3.11. Team comparison table and cumulative team points against the baseline.

3.3 Track Page

Track Page Description. The Track page (Figure 3.12) is designed as a circuit-centric view. It enables users to examine the technical characteristics of each circuit, identify the Grand Prix winner and notable lap records, analyse the relationship between starting grid positions and final race positions, and compare qualifying gaps to pole position. A Track slicer controls the circuit under analysis, and all visuals on the page update accordingly.

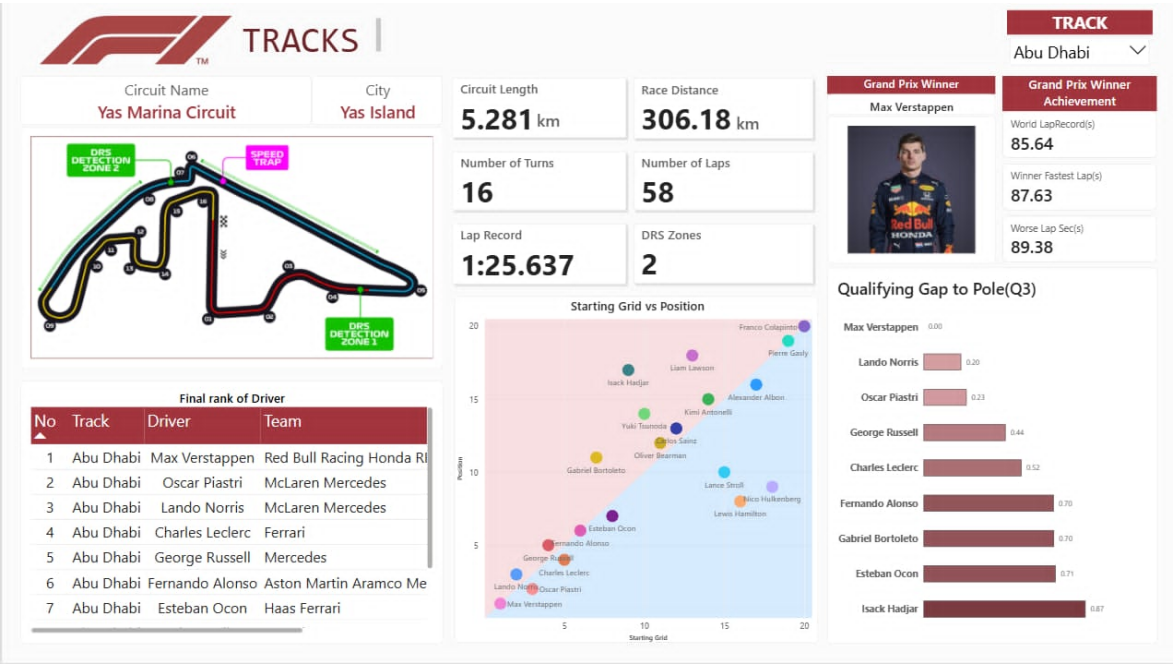


Figure 3.12. Track Page – Circuit profile, qualifying performance, and race results.

The purpose of this page is to present the technical profile of a circuit, including its length, race distance, number of turns, number of laps, lap record, and DRS zones. It also shows the Grand Prix winner at the selected circuit together with notable lap records, analyses how starting grid positions relate to final race outcomes, examines qualifying gaps to pole position in Q3, and provides a detailed final classification table for the race.

The main interaction on this page is the Track slicer, which is configured as a single-select control and filters all visuals on the page. In addition, a Position slicer can be used to restrict charts and tables to selected finishing ranges, such as the Top 3 or Top 10. A Driver filter at visual level is also applied to the scatter plot and the qualifying-gap bar chart when the analysis needs to focus on specific drivers.

Visual 1: Circuit Profile KPIs. This visual consists of multiple KPI cards and is used to provide a technical profile summary of the selected circuit.

A set of KPI cards is displayed for the selected circuit, including **Circuit Name** as text, **City** as text, **Circuit Length** as a numeric value such as 5.278 km, **Race Distance** as a numeric value such as 306.1 km, **Number of Turns** as an integer, **Number of Laps** as an integer, **Lap Record** as a time value such as 1:19.813, and **DRS Zones** as an integer. As KPI cards, these visuals do not require X-axis or Y-axis fields. The colour treatment remains neutral,

and each card is styled consistently so that the full set forms a circuit profile band. No legend is required. The Track slicer determines which circuit's KPI values are displayed. The main measures and fields include $\text{Circuit Length} = \text{Track Length (km)}$, $\text{Race Distance} = \text{Circuit Length} * \text{Number of Laps}$, $\text{Number of Turns} = \text{COUNT}(\text{Turns})$, $\text{Number of Laps} = \text{Scheduled Laps}$, $\text{Lap Record} = \text{MIN}(\text{Lap Time})$ for the circuit, and $\text{DRS Zones} = \text{COUNT}(\text{DRS Zone})$.

These KPI cards answer the question of what the key technical characteristics of the selected circuit are, including length, distance, number of turns, laps, lap record, and DRS zones.

Each of these attributes is a single scalar value, so KPI cards provide a concise and appropriate presentation without the need for axes or additional chart elements. This arrangement also allows users to build a mental model of the circuit before moving to more complex visuals such as the scatter plot or qualifying-gap chart.

The circuit KPI band is shown in Figure 3.13.

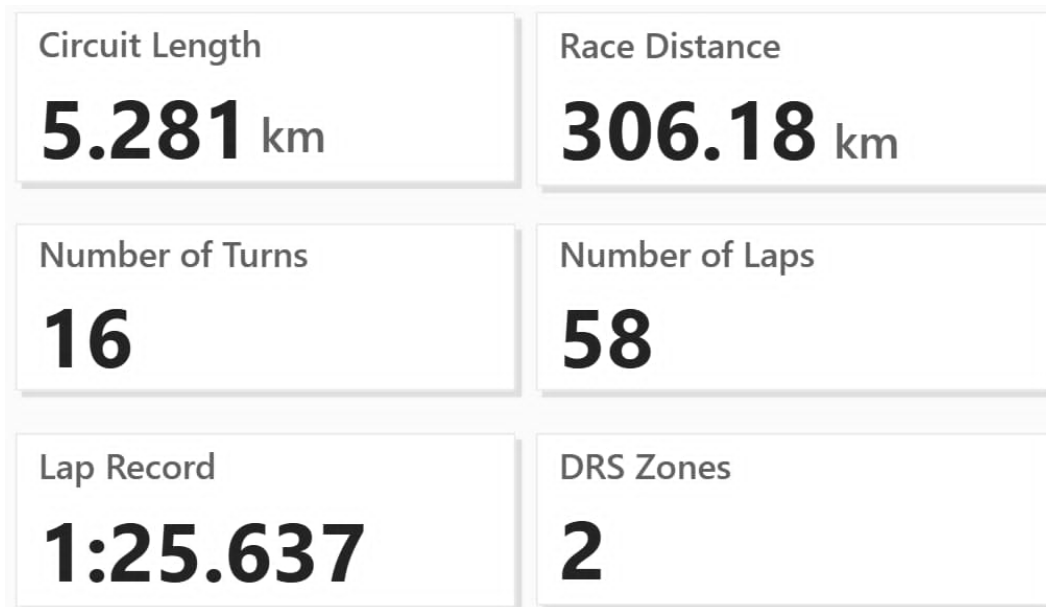


Figure 3.13. Circuit profile KPI cards for the selected track.

Visual 2: Track Map Image. This visual is implemented as an image visual and is used to provide a spatial layout illustration and contextual map.

The content of this visual is the graphical layout of the circuit, which may optionally include corners, DRS zones, the speed trap, and detection zones. Because it is an image rather than plotted data, X-axis and Y-axis fields are not applicable. Colour and size are inherent to the map image itself, and no additional data is encoded visually. Any legend that is needed may be embedded directly inside the map graphic. The Track slicer updates the displayed image when the selected circuit changes. The main field used is `DimTrack[Track Map Image URL]` or an equivalent field.

The image answers the question of what the circuit layout looks like and where the key zones, such as DRS areas and the speed trap, are located.

A static image is the most direct method for presenting circuit geometry, as this information would be difficult to communicate effectively through purely numeric charts. It also

provides important context for understanding why some circuits are more favourable to over-taking than others.

The map-based circuit profile is shown in Figure 3.14.

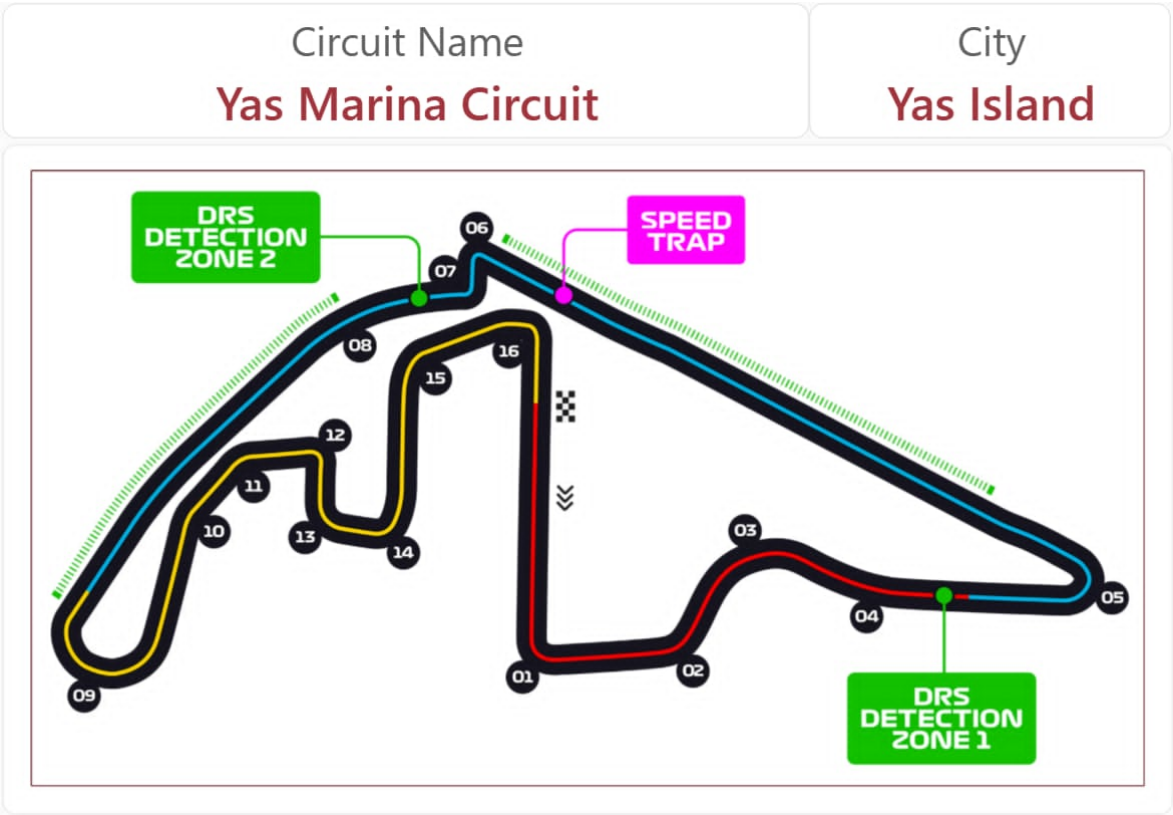


Figure 3.14. Track map and location profile for the selected circuit.

Visual 3: Grand Prix Winner and Achievement Metrics. This visual combines a profile card with a multi-row card and is used to highlight the race winner and associated lap records.

The winner card contains the Grand Prix winner’s name, team, and driver photo. Alongside it, the achievement card in multi-row format displays World Lap Record(s) for the track, Winner Fastest Lap(s), and Worse Lap Sec(s) = Worst Lap – Best Lap. No axes are required for either component. The colour palette remains neutral, although the winner card may use a team-colour accent. A legend is not necessary. The Track slicer determines which race result is displayed, while the Position slicer can be used to isolate the winner by restricting the data to position=1 when calculating certain metrics. The main measures are Grand Prix Winner = Driver with Position = 1, World Lap Record(s) = MIN(All Recorded Lap Times), Winner Fastest Lap(s), and Worse Lap Sec(s) = Worst Lap Time – Best Lap Time.

This visual answers the question of who won the race at the selected circuit and which notable performance metrics, including lap records, fastest laps, and lap spread, are associated with that track.

A profile card is well suited to presenting the winner and team in a clear and narrative-oriented format. The accompanying multi-row card allows several important lap metrics to be shown together without requiring separate small charts for each measure.

The winner and achievement cards are shown in Figure 3.15.

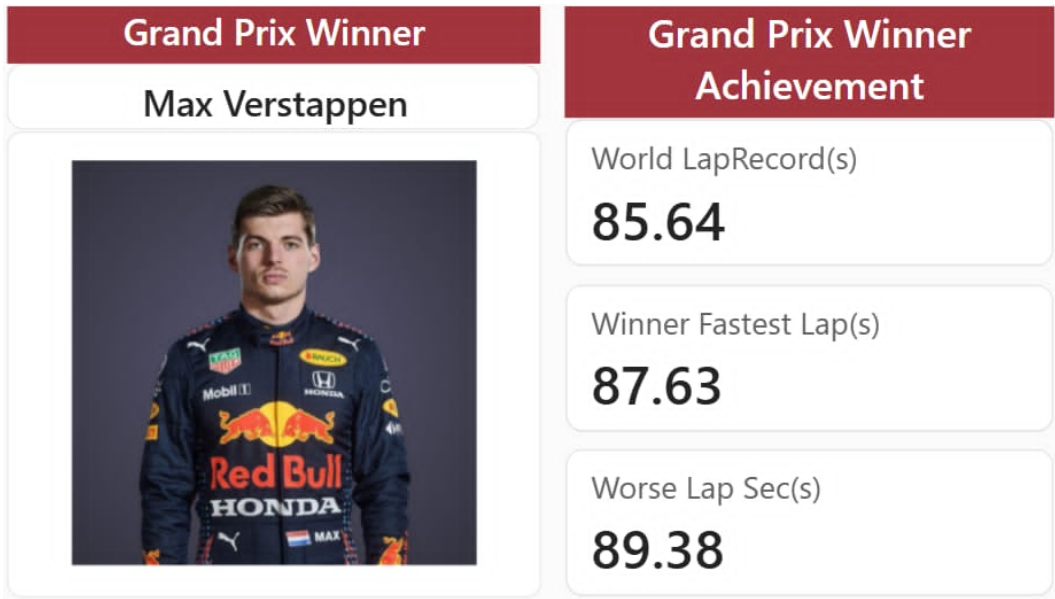


Figure 3.15. Grand Prix winner profile and achievement metrics for the selected circuit.

Visual 4: Starting Grid vs Finish Position (Race Scatter Plot). This visual is a scatter plot designed for relationship analysis between qualifying position and race result, as well as for detecting outliers.

The X-axis represents Starting Grid, while the Y-axis represents Finish Position. Colour is used to distinguish either Driver or Team through a qualitative hue scale. Marker size is typically constant, meaning it is not used as an additional data channel. A legend may identify drivers or teams, while specific names can also be displayed in tooltips. The Track slicer restricts the visual to the selected race, the Position slicer can limit the result to selected finishing ranges such as the Top 10, and a Driver visual-level filter can be applied when focusing on selected drivers. The main fields and measures are Starting Grid = Grid Position, Finish Position = Race Position, and Position Change = Starting Grid - Finish Position.

The scatter plot addresses which drivers gained or lost the most positions relative to their starting grid and how strong the relationship is between starting position and final result at the selected circuit.

Scatter plots are appropriate for examining the relationship between two numeric variables and are particularly useful for identifying outliers, such as unusually large gains or losses in position. In comparison with a table of starting and finishing positions, the scatter plot makes the overall pattern and spread visible much more quickly.

The grid-to-finish relationship is shown in Figure 3.16.

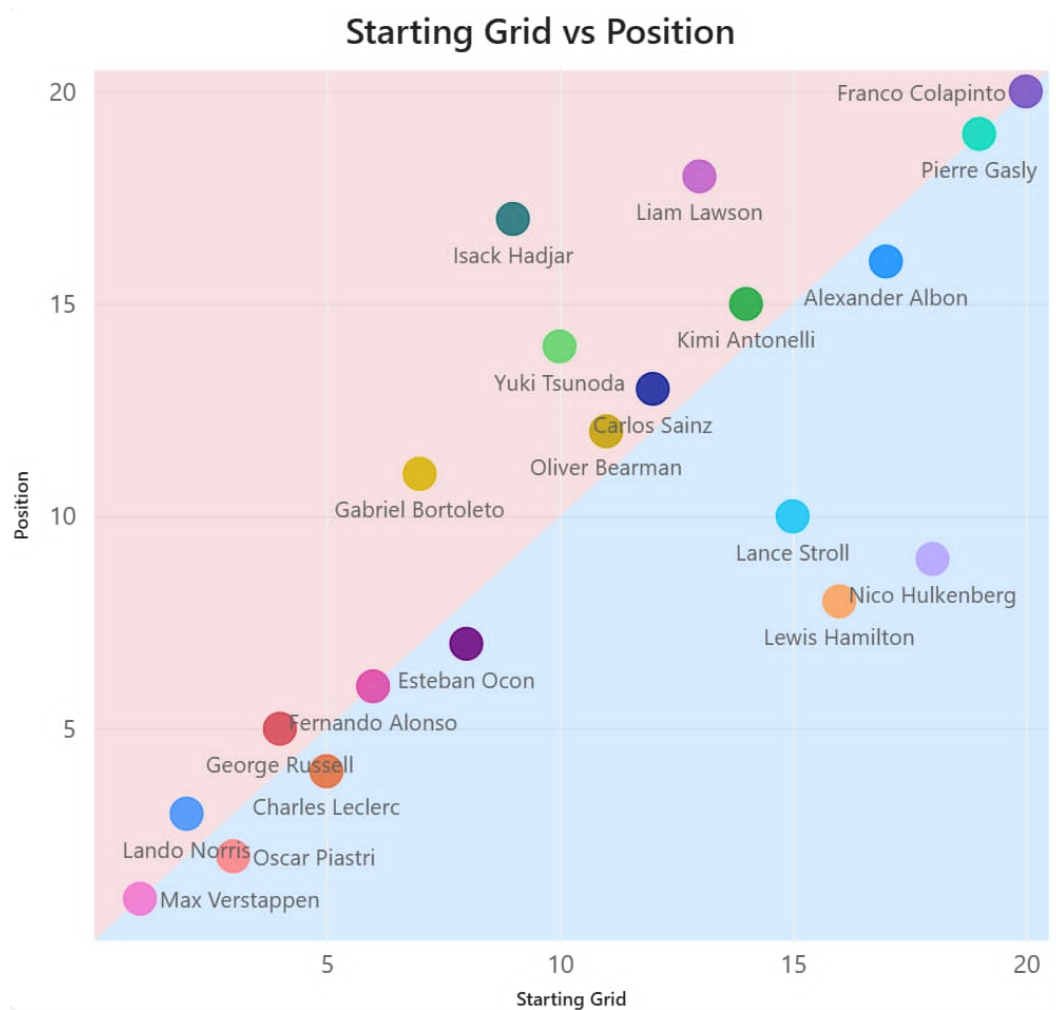


Figure 3.16. Track-level scatter plot comparing starting grid and finishing position.

Visual 5: Qualifying Gap to Pole (Q3) (Horizontal Bar Chart). This visual is implemented as a horizontal bar chart and is used for difference-from-reference comparison and ranking by qualifying gap.

The X-axis shows Q3 Delta, which represents the time difference to pole position in seconds, while the Y-axis lists Driver. Colour may remain neutral or be assigned by team, although a neutral palette may be preferred when the intention is to emphasise the size of the gaps. Size is not encoded beyond bar length. A legend may be included if team colours are used. The Track slicer determines which circuit's Q3 session is shown, a Driver visual-level filter can restrict the chart to selected drivers, and the Position slicer may be used to focus on groups such as the top-10 qualifiers. The main measures are $Q3\ Delta = \text{Driver } Q3\ Time - \text{Pole Time}$ and $Q3\ \% \text{ Gap} = \text{DIVIDE}(Q3\ Delta, \text{Pole Time}) * 100$.

This chart answers how far each driver was from pole position in Q3 and whether qualifying at the selected circuit was closely contested or widely spread.

A bar chart supports direct comparison of time gaps, with the shortest bar representing the driver closest to pole. Sorting the bars by gap makes it easier to identify the most competitive drivers and to assess the overall competitiveness of the qualifying session.

The Q3 gap chart is shown in Figure 3.17.

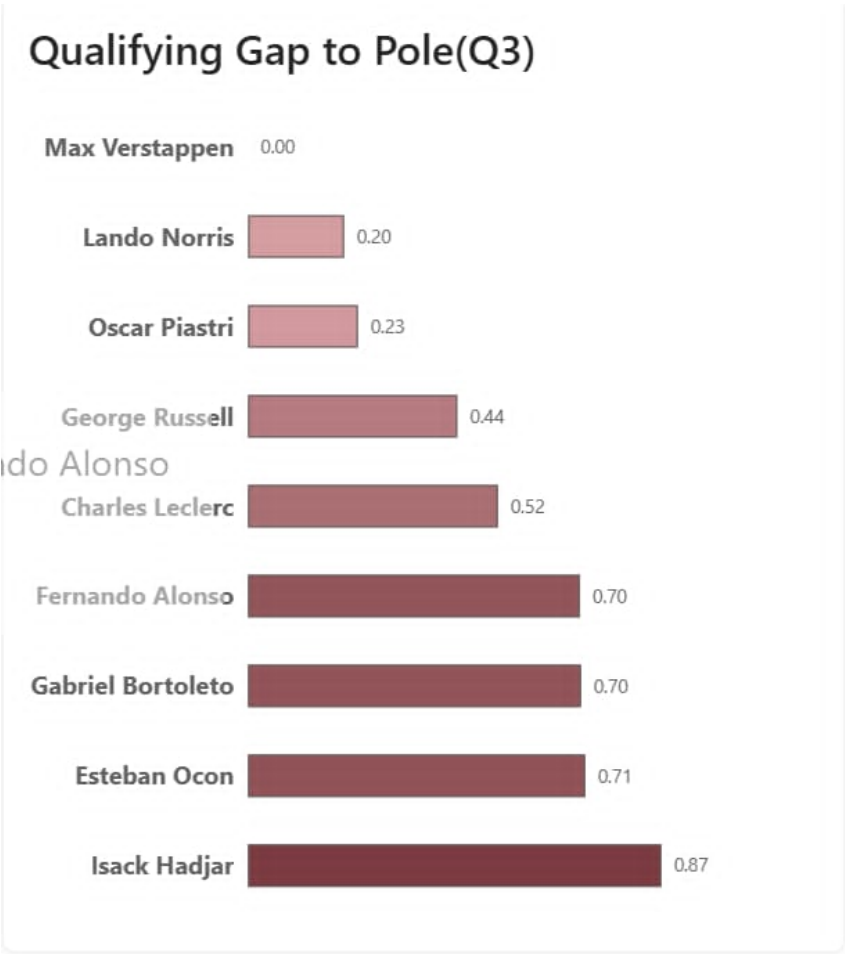


Figure 3.17. Q3 qualifying gap to pole shown as a horizontal bar chart.

Visual 6: Final Rank of Driver (Race Results Table). This visual takes the form of a table and is used for detailed classification listing and exact lookup.

The table includes the columns `No` for position number, `Track` for circuit name, `Driver`, `Team`, and additional columns where available, such as points scored or race status. Since this is a tabular layout, X-axis and Y-axis fields are not applicable. Colour usage is minimal, although it may be applied to highlight the winner or selected drivers. No legend is necessary. The `Track` slicer filters the table to the race at the selected circuit, and the `Position` slicer can be used to limit the rows to top-N finishers. The main fields and measures include `Position`, `Driver`, `Team`, and optional race-specific metrics.

The table provides the final finishing order at the selected circuit and shows which drivers and teams occupied each position.

A table is the most practical format for presenting the complete race classification when exact positions, driver names, and team names are required. It complements the scatter plot and Q3 gap chart by supplying precise details after broader visual patterns have already been explored.

3.4 Drivers Page

Drivers Page Description. The Drivers page (Figure 3.18) is designed as a driver-centric view of the season. It enables users to focus on an individual driver and examine their total points, championship rank, podium finishes, DNFs, and profile metrics, while also supporting direct comparison with their teammate. In addition, the page is intended to analyse the driver’s performance across rounds, including finishing positions and grid-to-finish gains or losses. A Driver slicer, often constrained by a Team slicer, determines which driver is being analysed, and all visuals on the page update together as coordinated views.

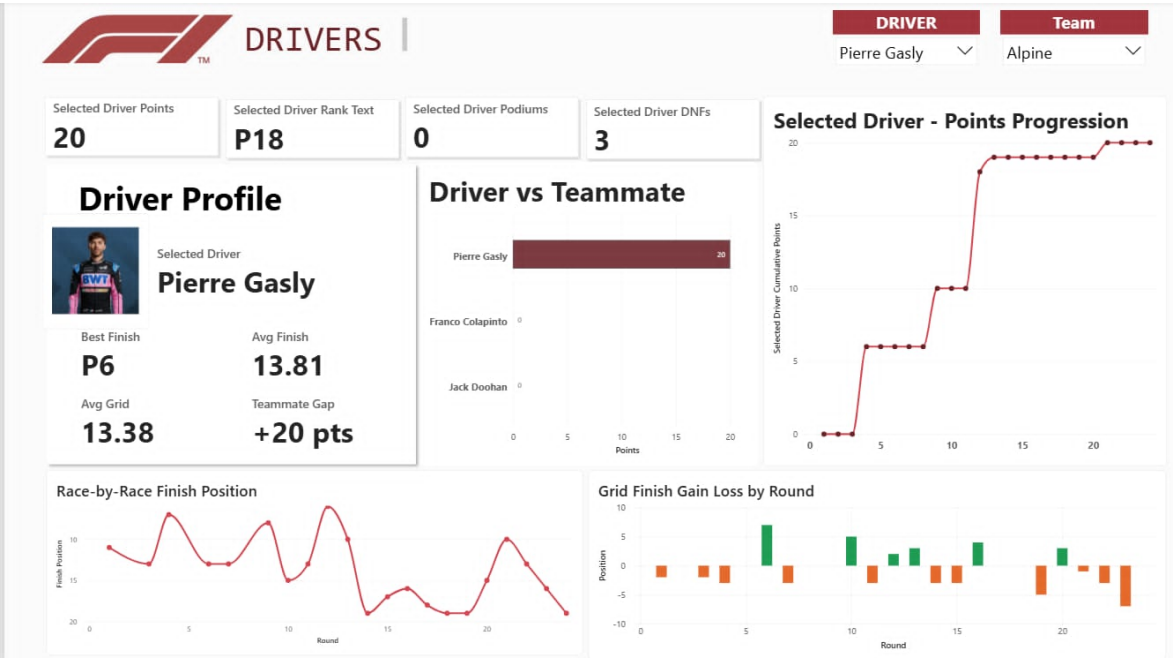


Figure 3.18. Drivers Page – Individual driver performance, consistency, and teammate comparison.

The purpose of this page is to provide a detailed performance summary for a selected driver, including points, rank, podiums, and DNFs. It also presents a profile of the driver through typical starting and finishing positions, best result, and gap to their teammate. Beyond the summary layer, the page allows direct comparison of points against the teammate, visualises how cumulative points and race results evolve round by round, and analyses gains or losses between starting grid and final finishing position on a race-by-race basis.

The Team slicer restricts the driver list to those belonging to the selected team. The Driver slicer is then used to choose a single driver, and all visuals on the page update according to that selection. Where available, a Round filter can further restrict round-by-round charts to a selected subset of the season, such as Rounds 1 to 10.

Visual 1: Driver KPI Band (Points, Rank, Podiums, DNFs). This visual is implemented as a set of KPI cards and is used to provide a driver-level summary.

Each KPI card displays one key metric for the selected driver. These include **Selected Driver Points**, representing the driver’s total points in the season; **Selected Driver Rank**, indicating their current championship position such as P1 or P5; **Selected Driver Podiums**, showing the number of podium finishes; and **Selected Driver DNFs**, showing the number of

DNFs. Since the visual is composed of KPI cards, X-axis and Y-axis fields are not applicable. The colour scheme is neutral, although an accent may be applied to rank or DNFs where appropriate, and no legend is required. The Team slicer limits which drivers are available for selection, while the Driver slicer determines whose KPI values are displayed. The main measures are `Selected Driver Points`, which returns total season points for the selected driver; `Selected Driver Rank Text`, which produces a label such as P1 or P2 based on points and tie-break logic; `Selected Driver Podiums`, which counts finishes in positions 1 to 3; and `Selected Driver DNFs`, which counts races with status equal to DNF.

These KPI cards answer how many points the selected driver has scored, what their championship rank is, how many podiums they have achieved, and how many DNFs they recorded.

These four indicators form the core summary of driver performance and are best presented as simple KPIs for immediate reading. A more complex chart would introduce unnecessary visual overhead without offering additional value for these basic metrics.

The selected-driver KPI band is shown in Figure 3.19.



Figure 3.19. Driver KPI band showing points, rank, podiums, and DNFs for the selected driver.

Visual 2: Driver Profile Card (Best Finish, Avg Grid, Avg Finish, Teammate Gap). This visual is implemented as a composite card and is used to summarise driver profile information together with the comparative gap to their teammate.

The card contains the driver’s name and photo, together with four key profile metrics. These are **Best Finish**, which records the best race result such as P1 or P2; **Avg Grid**, representing average starting grid position; **Avg Finish**, representing average finishing position; and **Teammate Gap**, which is defined as the points difference between the selected driver and their teammate. As a composite card, it does not require X-axis or Y-axis fields. Team-colour accents may be used, while a legend is not needed. The Team slicer defines the teammate used for comparison, and the Driver slicer determines whose profile is displayed. The main measures are `Best Finish`, defined as the minimum finishing position for the selected driver; `Avg Grid`, calculated as the average grid position across races, excluding DNS or DQ if the model is defined in that way; `Avg Finish`, calculated as the average finishing position across races, excluding DNF, DNS, or DSQ depending on the DAX logic; and `Teammate Gap = Selected Driver Points - Teammate Points`.

This visual answers what the overall performance profile of the selected driver looks like in terms of starting position, finishing position, best result, and the points difference relative to their teammate.

A composite card allows several profile indicators to be presented together in one place, which helps maintain readability without requiring multiple separate charts. It also establishes a clear narrative context before the user moves to more detailed comparison and time-series visuals.

The driver profile card is shown in Figure 3.20.

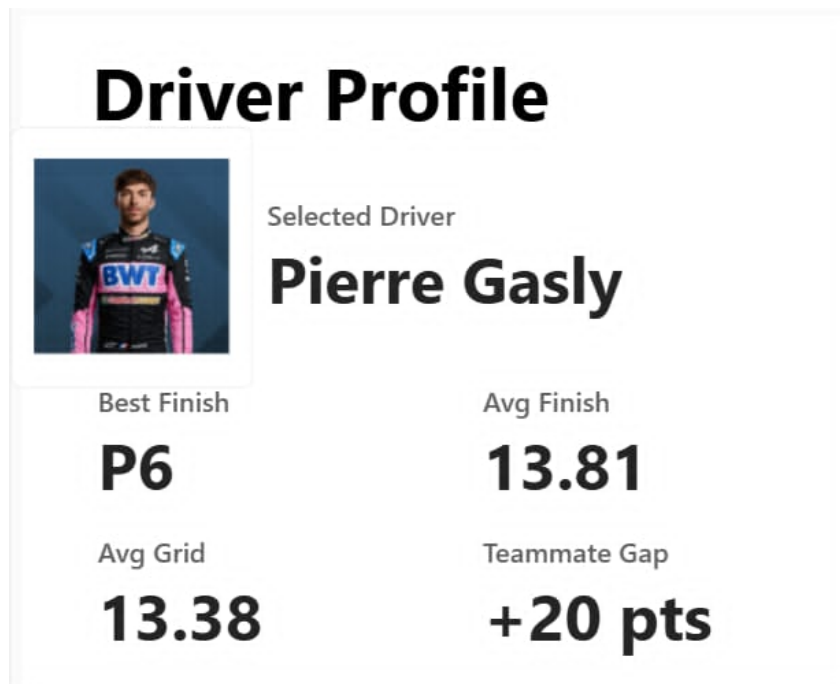


Figure 3.20. Driver profile card with image, best finish, average grid, average finish, and teammate gap.

Visual 3: Driver vs Teammate (Horizontal Bar Chart). This visual is implemented as a horizontal bar chart and is used for direct intra-team comparison between two drivers.

The X-axis represents `Total Points`, while the Y-axis lists the two drivers, namely the selected driver and their teammate. Colour is used to distinguish the selected driver from the teammate, with one accent colour applied to the selected driver and a neutral colour applied to the teammate. Size is not used beyond bar length. A simple legend may be included to map colours to “Selected Driver” and “Teammate” where necessary. The Team slicer determines which driver pair appears in the chart, and the Driver slicer identifies which of the two bars should be treated as the selected driver. The main measure is `Total Points` for each driver in the team.

The chart answers whether the selected driver is scoring more or fewer points than their teammate and by what margin.

A horizontal two-bar chart provides a direct and easily interpretable comparison between teammates by using bar length on a common scale. It also aligns closely with the teammate gap presented in the profile card and visually reinforces that metric.

The teammate comparison is shown in Figure 3.21.

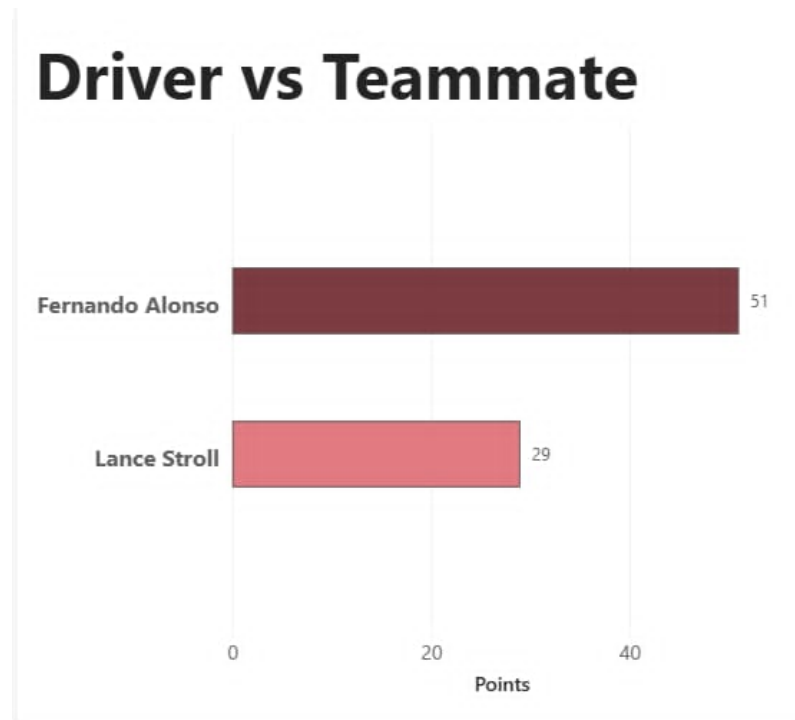


Figure 3.21. Driver versus teammate point comparison using a horizontal bar chart.

Visual 4: Round-by-Round Performance (Cumulative Points, Finish Position, Grid-Finish Gain/Loss). This section combines multiple chart types, specifically a line chart for cumulative points, a line chart for finish position by round, and a column chart for grid-to-finish gain or loss. Together, these visuals support time-series trend analysis, race-by-race performance tracking, and position-change analysis.

(a) Cumulative Points Progression (line chart)

In the cumulative points chart, the X-axis represents Round and the Y-axis represents Selected Driver Cumulative Points. A single line colour is used for the selected driver, and size is not employed as an additional encoding channel. Since only one driver is shown at a time, a legend is usually not necessary. The Driver slicer selects the driver whose progression is displayed, while the Round filter, if present, narrows the time window to a selected portion of the season, such as Rounds 1 to 10. The main measure is Selected Driver Cumulative Points, defined as the running total of the driver's points ordered by round.

(b) Race-by-Race Finish Position (line chart)

In the finish position chart, the X-axis is again Round, while the Y-axis represents Finish Position, where 1 indicates a win and higher numbers indicate lower finishing places. A single line is used for the selected driver, and no size encoding is applied. Because the visual focuses on one driver, a legend is not required. The Driver slicer selects the driver shown, and the Round filter limits the visible rounds where necessary. The main field or measure is Finish Position, representing the finishing position in each race, with DNF potentially encoded separately depending on the data model.

(c) Grid-Finish Gain/Loss (column chart)

In the gain/loss chart, the X-axis represents Round and the Y-axis shows Grid Finish Gain Loss. Colour is divided into two categories, with one colour assigned to gains rel-

ative to the starting grid and another assigned to losses. Size is not used beyond the height of the columns, and the legend indicates which colour corresponds to gains and which corresponds to losses. The Driver slicer determines which driver's race-by-race gain or loss is displayed, while the Round filter restricts the set of races where needed. The main measure is $\text{Grid Finish Gain Loss} = \text{Grid Position} - \text{Finish Position}$, where the sign convention in the model determines whether negative values indicate gained positions and positive values indicate lost positions.

The cumulative points chart answers how the selected driver's total points evolve over the season and identifies the rounds in which points are gained rapidly or remain unchanged. The finish position chart addresses how consistent the driver's race finishing results are across rounds and whether there are clear peaks, such as wins, or troughs, such as poor finishes or DNFs. The grid–finish gain/loss chart shows whether the driver typically gains positions, loses positions, or maintains their grid slot in each race.

The cumulative points line chart provides a clear season-long trend and makes strong or weak scoring phases easy to identify. The finish position line chart complements this by showing the race outcomes that shape the cumulative curve. The gain/loss column chart focuses on racecraft and makes it easier to see where the driver performed above or below expectation relative to starting position. Using the same X-axis, namely Round, across all three visuals also helps users align events mentally and interpret the season as a coherent timeline.

The cumulative points progression is shown in Figure 3.22.

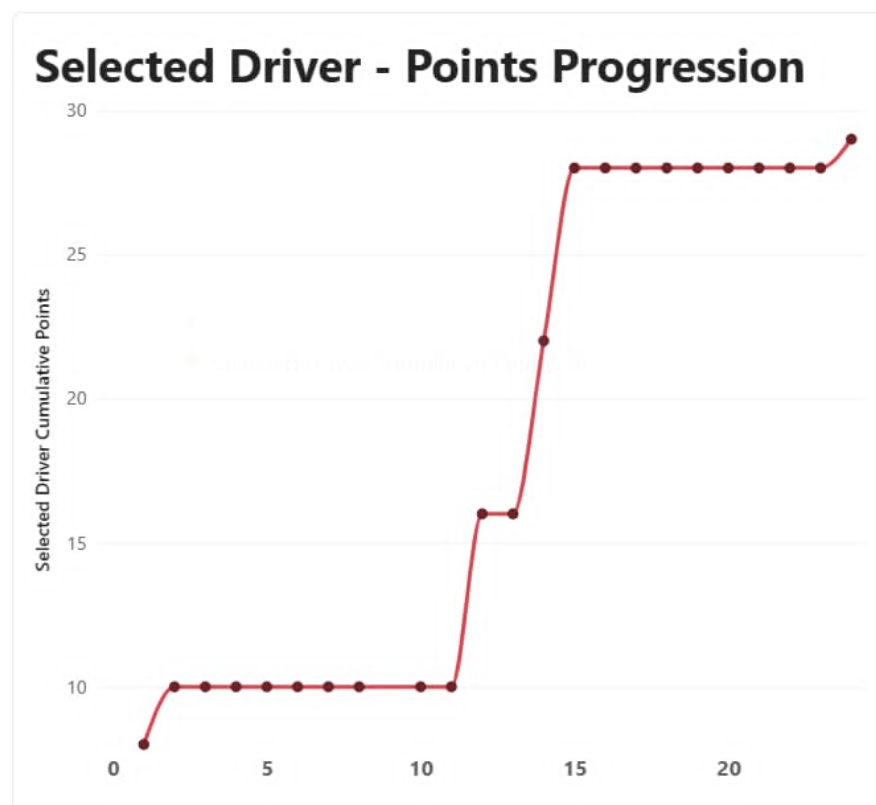


Figure 3.22. Selected driver cumulative points progression across race rounds.

The race-by-race finish trend is shown in Figure 3.23.

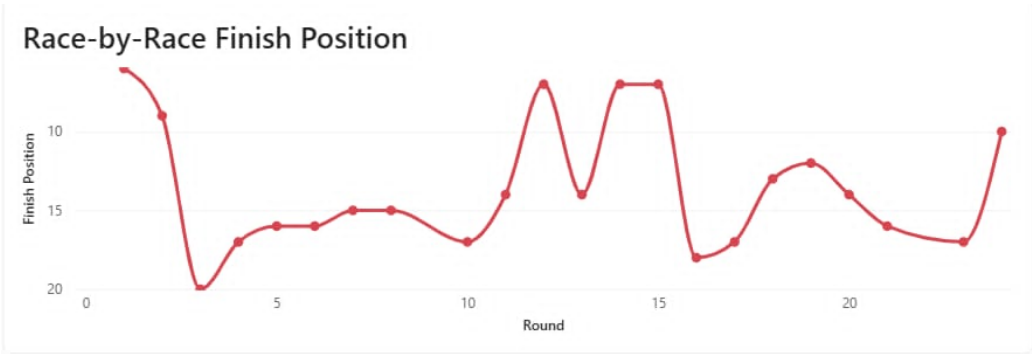


Figure 3.23. Race-by-race finishing position for the selected driver.

The grid-to-finish gain/loss chart is shown in Figure 3.24.

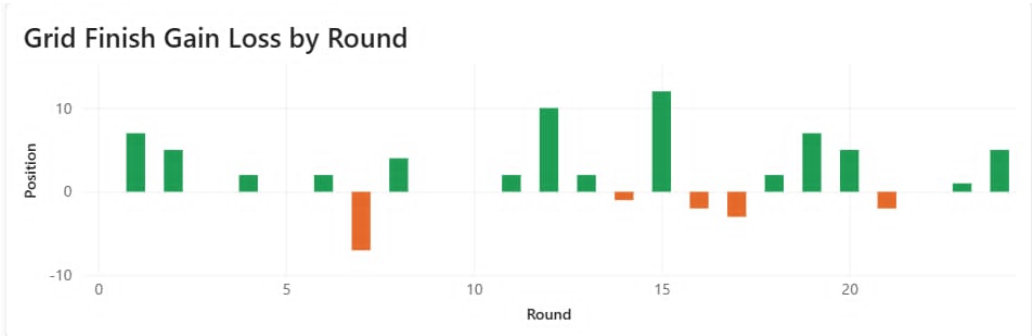


Figure 3.24. Grid-to-finish gain and loss by round for the selected driver.

Technique Application

4.1 Course Chapter 1: Dataset Choice and Role of Visualization

4.1.1 Techniques / Principles Applied

- Clarifying the analytical questions and the target audience before designing visualizations (Who / What / How).
- Using visualizations primarily for exploratory analysis with selected elements of explanatory storytelling.
- Choosing a dataset that naturally supports multiple perspectives (drivers, teams, tracks) and benefits from interactive filtering.

4.1.2 How Applied in the Dashboard

The Formula 1 2025 season dataset was chosen because it provides rich, multi-dimensional data that are difficult to understand from raw tables alone. Typical questions for the intended audience (F1 fans, students, and analysts) include comparing driver performance across races, understanding team competitiveness over the season, and examining how circuit characteristics relate to race results. These questions require integrating information from several tables and viewing it through coordinated visualizations rather than isolated numbers.

The dashboard is designed mainly for exploratory analysis: users can select a team, driver, or track and immediately see how related charts update across pages. At the same time, key summary tiles and titles (for example, total points, best finish, or average grid position) play an explanatory role by highlighting important takeaways without requiring detailed exploration. Thus, the visualization acts both as a lens for interactive exploration and as a communication tool that summarises the main patterns of the 2025 championship. This overall exploratory structure can be seen across the four dashboard pages shown in Figures 3.1, 3.8, 3.12, and 3.18.

4.1.3 Notes / Adjustments

The project focuses on a single season instead of multiple historical seasons in order to keep the dashboard concise and avoid overwhelming users with too many filters. This scope allows the visual design to concentrate on depth of analysis (drivers, teams, tracks) rather than temporal comparisons across many years. If additional seasons were included, more pages or small-multiple panels would be needed to maintain clarity.

4.2 Course Chapter 2: Data Types and Visual Encodings

4.2.1 Techniques / Principles Applied

- Distinguishing nominal, ordinal, and quantitative variables to choose suitable visual encodings.
- Using position on a common scale and bar length as primary channels for quantitative comparison, which is consistent with graphical perception theory.
- Using color hue to encode nominal categories such as teams and drivers, and color value (intensity) to reflect quantitative magnitude where appropriate.
- Avoiding angle- and area-based encodings (e.g., complex pie charts, bubbles) for tasks that require precise comparison.

4.2.2 How Applied in the Dashboard

Across the dashboard, quantitative measures such as race points, cumulative points, average grid and finish positions, podium rates, and qualifying gaps are encoded using position and length. For example, on the *Overview* page, the *Race Points by Top 5 Driver* chart displays each driver's total points as bar length on a zero-based axis, supporting accurate comparison of magnitudes. Similarly, the *Race Points by Team* bar chart encodes team-level points using bar length, and the stacked *Constructor Standings & Driver Contributions* visual shows both total constructor points and each driver's contribution via stacked bar segments. These encodings are illustrated in Figures 3.3, 3.5, and 3.6.

Nominal variables are primarily encoded by color hue and textual labels. Teams are assigned distinct but consistent colors (e.g., Ferrari red, McLaren orange, Mercedes silver), which appear in the *Overview* and *Teams* pages and are reused in driver-level charts so that users can immediately recognise the same team across pages. Driver identities are encoded by category labels and, where space allows, by portraits (e.g., in the *Drivers* and *Track* pages) rather than by complex combinations of color and shape.

Ordinal variables such as starting grid position and finishing position are represented using ordered axes. On the *Drivers* page, the *Race-by-Race Finish Position* line chart places race rounds on the horizontal axis and finish positions on the vertical axis, preserving the natural order of positions and allowing users to track consistency across the season. On the *Track* page, the *Starting Grid vs Position* scatter plot maps grid position to the x-axis and final position to the y-axis, so that movement above or below the diagonal reflects ordinal gains or losses during the race.

Some derived measures combine ordinal and quantitative information. For instance, the *Grid Finish Gain/Loss* column chart on the *Drivers* page uses the measure Finish Position – Grid Position: negative values correspond to drivers gaining positions, positive values to losing positions, and zero to holding position. These values are encoded by position on the vertical axis and reinforced by diverging colors (e.g., green for gains, orange for losses), making the direction of change immediately apparent, as shown in Figure 3.24.

4.2.3 Notes / Adjustments

Pie charts and bubble charts were deliberately avoided even though some analyses involve proportions (e.g., team share of total points or distribution of fastest laps). In line with the course notes, position and length are more accurate channels than angle or area for comparing values, especially when differences are small. For the same reason, the *Fastest Lap Leaders* visual on the *Overview* page uses a treemap with limited categories rather than a multi-slice pie chart, and stacked bars are preferred for showing driver contributions within teams.

4.3 Course Chapter 3: Pre-attentive Processing and Gestalt Principles

4.3.1 Techniques / Principles Applied

- Using pre-attentive attributes such as color, size, and orientation to draw instant attention to key elements.
- Applying Gestalt principles of proximity, similarity, and common region to organise charts and KPIs on each page.
- Using magnitude estimation primarily through length and position, and avoiding misleading size encodings.

4.3.2 How Applied in the Dashboard

On each dashboard page, pre-attentive color is used to highlight the currently selected team, driver, or track. For example, in the *Drivers* page, the selected driver is shown with a saturated accent color in the profile card and in the *Driver vs Teammate* bar chart, while other elements use neutral tones. This design allows users to identify the focus of the analysis within a fraction of a second without reading all labels, as illustrated by the driver profile and teammate comparison in Figures 3.20 and 3.21.

Gestalt principles guide the layout of components. On every page, top-level KPI cards (such as total points, average finish, number of podiums, or circuit length) are grouped in a single row at the top, separated from detailed charts below. This consistent placement leverages proximity and common region so that users perceive the KPIs as a coherent summary layer above the detailed analysis layer. Examples of this KPI grouping are shown in Figures 3.2, 3.13, and 3.19.

Similarity is used by keeping fonts, colors, and border styles consistent for related elements. For instance, all driver-related tiles and charts share the same header color and typography, reinforcing the perception that they belong to one analytical group. Conversely, visual separation between the *Overview*, *Teams*, *Track*, and *Drivers* pages prevents users from confusing the context of filters and selections.

4.3.3 Notes / Adjustments

The dashboard intentionally avoids using too many pre-attentive channels at once on a single chart (for example, combining color, size, and shape simultaneously). While adding more encodings could increase the amount of information displayed, it would also increase cognitive load and make it harder to interpret patterns. Instead, important points or anomalies are emphasised using annotations and selective color, following the storytelling guidelines from the course.

4.4 Course Chapter 4: Chart Types and Analytical Tasks

4.4.1 Techniques / Principles Applied

- Matching chart types to analytical tasks such as comparison, ranking, trend analysis, and relationship exploration.
- Preferring bar and line charts for quantitative comparison in order to leverage position and length as highly accurate channels.
- Using scatter plots to study relationships between continuous variables (e.g., grid vs finish position) and treemaps for simple part-of-whole questions.
- Avoiding 3D charts and complex pie charts that could violate proportional ink and hinder accurate perception.

4.4.2 How Applied in the Dashboard

On the *Overview* page, several chart types address high-level comparison tasks. The *Race Points by Top 5 Driver* column chart allows users to compare the top drivers' total points directly, while the *Race Points by Team* horizontal bar chart summarises constructor performance across the season. The stacked *Constructor Standings & Driver Contributions* visual combines a ranking of teams by total points with a view of how points are split between drivers within each team, revealing whether a team relies heavily on one driver or benefits from balanced contributions. These charts are shown in Figures 3.3, 3.5, and 3.6.

The *Teams* page focuses on team-level trends and context. A set of KPI cards summarises each team's average points per race, podium rate, and historical number of championships, providing a quick performance snapshot. The *Driver Contribution of Team over Track* bar chart shows how points are distributed between drivers across circuits, and a line chart of *Cumulative Team Points vs Baseline* plots each team's running total of points against a notional baseline, making it easy to see where the team over- or under-performs expectations. These applications are shown in Figures 3.10 and 3.11.

The *Track* page employs both summary and analytical charts tailored to circuit-level questions. After basic circuit statistics are shown via card visuals (length, distance, turns, laps, lap record, DRS zones), a scatter plot of *Starting Grid vs Position* displays the relationship between starting and finishing positions for all drivers at the selected track. A horizontal bar chart of *Qualifying Gap to Pole (Q3)* shows each driver's time delta relative to pole position, which is a natural use case for ordered bars and supports questions about how competitive

the qualifying session was. The corresponding visuals are shown in Figures 3.13, 3.16, and 3.17.

The *Drivers* page combines trend, comparison, and detailed performance charts. KPI cards summarise total points, current championship rank, number of podiums, and number of DNFs for the selected driver. The *Driver vs Teammate* bar chart compares total points between the selected driver and their teammate, while the *Selected Driver – Points Progression* line chart shows cumulative points by round, highlighting phases of rapid improvement or stagnation. A line chart of *Race-by-Race Finish Position* and a diverging column chart of *Grid Finish Gain/Loss* provide race-level diagnostics for consistency and racecraft. These driver-level applications are shown in Figures 3.21, 3.22, 3.23, and 3.24.

4.4.3 Notes / Adjustments

Some chart variants were considered but not included in order to keep the dashboard coherent and readable. For instance, position-change analysis on the *Track* page could be shown as a slopegraph between grid and finish positions, but the scatter plot was preferred because it scales better with the number of drivers and supports visual clustering. Similarly, while lap-time distributions could be visualised using histograms or density plots, they were omitted from the current version to avoid overloading the *Track* page and to keep the focus on standings, qualifying gaps, and position changes.

4.5 Course Chapter 6: Proportional Ink, Color, and Layout

4.5.1 Techniques / Principles Applied

- Applying the proportional ink principle: the size of visual elements is proportional to the underlying data values.
- Ensuring bar charts start from zero to avoid exaggerating differences.
- Maximising the data–ink ratio by removing unnecessary borders, gridlines, and decorations.
- Using appropriate color schemes: qualitative palettes for nominal data and sequential palettes for quantitative intensity, while staying colorblind–friendly.
- Organising the dashboard into multiple coordinated pages (small multiples) instead of overcrowding a single view.

4.5.2 How Applied in the Dashboard

All bar charts in the dashboard are designed with a zero-based value axis. This guarantees that bar length is directly proportional to the corresponding value, preventing misleading visual exaggeration of small differences in points or counts. Similarly, line charts are scaled to show trends clearly without distorting relative changes. Examples include the ranking and trend charts in Figures 3.5, 3.22, and 3.11.

Unnecessary non-data ink is minimised. Axis lines and gridlines are kept light, backgrounds are mostly white, and chart borders are removed unless they are needed to separate panels. This increases the visual prominence of the actual data marks (bars, points, and lines) and follows the data-ink ratio recommendations from the course

Color is used systematically: qualitative palettes with distinct hues encode teams or drivers, while sequential palettes encode quantitative measures such as performance ratings or gap values. The palette avoids red-green combinations and rainbow schemes, making the dashboard accessible for users with color vision deficiencies. Within each page, one accent color is reserved for the selected entity, and less important elements are rendered in neutral grey tones. This can be seen in the team and driver views in Figures 3.10 and 3.20.

Instead of putting all information into one dense page, the dashboard is split into four coordinated pages: *Overview*, *Teams*, *Track*, and *Drivers*. This multi-panel design acts as a form of small multiples over analytical dimensions, allowing users to switch focus while preserving the same visual grammar and color coding. The result is a cleaner layout that supports comparison across pages without overwhelming the viewer.

4.5.3 Notes / Adjustments

In some charts, minor gridlines are retained to help users read approximate values without adding data labels to every bar. This is a trade-off between maximising data-ink and maintaining readability, and was judged acceptable given the limited number of series per chart. Heatmaps and 2D histograms were considered for dense data, but the number of points in each race was small enough that standard scatter plots remained readable without additional techniques for overplotting.

4.6 Course Chapter 8: Interaction Techniques

4.6.1 Techniques / Principles Applied

- Providing slicers as the main interaction mechanism so that users can filter the dashboard by key dimensions such as driver, team, and track.
- Using cross-filtering and cross-highlighting between charts on the same page to support coordinated views.
- Keeping the interaction model simple and consistent across pages, avoiding complex animations in favour of immediate, predictable responses.

4.6.2 How Applied in the Dashboard

On the *Overview* page, a *Driver* slicer and a *Team* slicer control the high-level summary of the season. Selecting a driver updates the champion card, the driver-focused bar charts (e.g., *Race Points by Top 5 Driver*), the *Grid Position vs Race Finish* scatter plot, and the *Fastest Lap Leaders* treemap so that the selected driver's performance is emphasised. Similarly, selecting a team filters the constructor champion card, the *Race Points by Team* bar chart, the stacked

Constructor Standings & Driver Contributions chart, and other visuals to that constructor, allowing users to read the entire season from one team's perspective.

On the *Teams* page, the main slicer is the *Team* selector, which is implemented as a single-select control. Choosing a team updates all visuals on the page: the logo and KPI cards, the *Team Performance by Track Characteristic* radar chart, the *Driver Contribution of Team over Track* chart, the team comparison table, and the cumulative points line chart. This design ensures that users can focus their analysis on one constructor at a time while preserving a consistent layout and visual encoding.

The *Track* page is centred around a *Track* slicer that filters the entire page to the selected circuit. When a track is chosen, the circuit map image, all circuit statistics cards, the Grand Prix winner profile, the *Starting Grid vs Position* scatter plot, the *Qualifying Gap to Pole* bar chart, and the final ranking table all update to reflect that race. Additional filters, such as position filters on the classification table or visual-level filters on specific drivers, allow more focused questions (for example, concentrating on top-10 finishers or a single driver's performance at that track) without changing the overall interaction pattern.

On the *Drivers* page, a *Driver* slicer and a *Team* slicer work together. The team slicer limits the list of drivers shown in the driver slicer to those belonging to the selected constructor, and the driver slicer then updates all visuals on the page, including the profile card, KPI tiles, teammate comparison bar chart, cumulative points line chart, finish-position line chart, and grid-gain/loss columns. This combination enables users to switch quickly between drivers and teams while always keeping the page focused on a single driver at a time.

4.6.3 Notes / Adjustments

Interactions are intentionally kept relatively simple: there are no animated transitions between pages or complex drill-through flows. This choice is consistent with the course recommendation that static, coordinated views (via slicers and cross-filtering) are often easier to compare than animated sequences. If the dashboard were extended for expert users, additional drill-through pages or dynamic tooltips could be introduced, but the current design already supports the main analytical tasks without overwhelming casual users.

4.7 Course Chapter 9: Storytelling with the Dashboard

4.7.1 Techniques / Principles Applied

- Defining the audience (Who), the main messages (What), and the delivery method (How) before designing the dashboard.
- Structuring the dashboard pages to follow a narrative flow from overview (macro) to detail (micro).
- Using descriptive titles, annotations, and visual hierarchy to guide attention and emphasise key insights.

4.7.2 How Applied in the Dashboard

The primary audience for the dashboard consists of F1 fans, students, and analysts who want to understand the 2025 season without writing queries or reading raw tables. The main messages focus on three questions: which teams and drivers performed best overall, how performance evolved across the season, and how circuit characteristics influenced race outcomes. These goals informed both the choice of visuals and the layout of the pages.

The narrative flow follows a macro-to-micro structure. Users start at the *Overview* page, which provides high-level summaries such as total points by team and driver, season standings, and basic season statistics. From there, they can move to the *Teams* page to explore how different drivers contribute to each constructor, then to the *Track* page to see how races played out at individual circuits, and finally to the *Drivers* page for in-depth profiles of specific drivers.

Titles and annotations are written to communicate insights rather than merely label axes. For example, instead of using a generic title such as “Cumulative Points”, a title may specify the selected entity, e.g. “Cumulative Points for Selected Driver over 2025 Rounds”, making the context explicit. Key anomalies, such as sudden drops in points progression or unusually large grid-to-finish gains, can be highlighted with short text notes in the charts to help users interpret the patterns quickly.

4.7.3 Notes / Adjustments

Because the dashboard is intended for both exploration and explanation, the narrative is not strictly linear. Users can enter the story from any page depending on their interests, and interactions update the context dynamically. However, the page order, consistent layout, and repeated color conventions provide enough structure so that first-time users can still follow a coherent storyline about the 2025 championship.

Conclusion, Limitations, and Future Work

5.1 Conclusion

This project demonstrates how core data visualization principles can be applied to design an interactive dashboard for the 2025 Formula 1 season in Power BI. By carefully matching data types to visual encodings, respecting graphical perception limits, and using interaction and storytelling techniques, the dashboard turns fragmented raw data into an integrated analytical tool. The same design process could be extended to other sports or domains where multi-dimensional performance data need to be communicated to a broad audience.

5.2 Limitations

While the dashboard provides valuable insights, it has several limitations regarding data utilization and technical execution. First, although our dataset spans from 2016 to 2026, this analysis focuses exclusively on the 2025 season, thereby missing out on long-term historical trends and future projections. Furthermore, we did not fully exploit all available data points; specifically, Sprint race information was omitted due to its structural complexity, which could have overcomplicated the current data model.

From a technical standpoint, the dashboard was constrained by our current proficiency in Power BI, which limited the variety and number of visualizations we could implement. As a result, some complex performance metrics could not be visualized as intended. Lastly, we acknowledge that the dashboard's aesthetic design and user interface have not yet reached a professional enterprise standard, as the focus was primarily on core data exploration rather than advanced visual storytelling..

5.3 Future Work

Building on the foundation of this 2025 analysis, future work will aim to unlock the full potential of our 2016–2026 dataset. By expanding the timeline, we can provide users with a deeper historical context that the current version lacks. We also intend to tackle the complexity of Sprint race data to ensure no performance points are overlooked.

Recognizing the current visual constraints, we plan to enhance our technical execution in Power BI, replacing summary-level charts with more precise, professional visualizations and interactive features. These upgrades will transform the dashboard from a basic exploratory tool into a high-fidelity analytical platform for F1 enthusiasts.

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